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Faculty of Information Technology



Course: Introduction to Software Engineering

UI PROTOTYPE REPORT

Workflow Management Module

Subject:	User Interface Prototype Report
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1 Introduction

This report presents the user interface prototype for the **Workflow Management** module of a virtual company workspace platform. In the overall product context, the platform supports authentication, workspace management, agent management, tools and integrations, workflow management, task orchestration, subscription management, workspace user management, and knowledge-base management.

Within that broader system, the Workflow Management module focuses on creating workflow definitions, editing workflow steps, assigning agents, validating workflow structure, executing workflows, monitoring execution status, and managing workflow-related settings.

The prototype is designed as a modern SaaS dashboard interface. This design choice is suitable for a workflow management system because users need a clear overview of workflows, structured tables for workflow definitions, visible status badges, and direct actions such as Create, Edit, Execute, and Monitor.

The purpose of this report is to document the UI screens, explain how each screen supports the module requirements, and demonstrate traceability between the prototype, functional requirements, and use cases.

2 Project and Module Context

The project aims to provide a company virtualization platform in which users can create a workspace, configure virtual employees, define workflows, connect tools, manage internal knowledge, and execute work through AI-based agents.

The Workflow Management module belongs to the process definition and execution layer of the platform. It is responsible for allowing users to design repeatable workflows and monitor their execution.

The module is responsible for six major operations:

- **Workflow overview:** showing overall workflow statistics and recent workflow activity.
- **Workflow listing:** displaying available workflows with their status, number of steps, and actions.
- **Workflow creation and editing:** allowing users to define workflow name, description, steps, dependencies, and assigned agents.
- **Workflow validation:** checking step order, assigned agents, dependencies, and estimated time before saving or publishing.
- **Execution monitoring:** showing execution status, history, and execution-related information.
- **Agent and settings management:** supporting agent assignment and workflow behavior configuration.

The UI prototype therefore does not attempt to cover the entire platform. It focuses on the user-facing screens needed for workflow overview, workflow creation, workflow editing, execution monitoring, agent management, and workflow configuration.

3 Relevant Requirements and Use Cases

3.1 Functional Requirements

The UI prototype is aligned with the following functional requirements of the Workflow Management module.

ID	Requirement	UI Coverage
FR-01	Workflow Overview	The Dashboard screen displays total workflows, running executions, completed executions, failed executions, recent workflows, and recent activity.
FR-02	Workflow List Management	The Workflows screen displays the workflow list with search, status badges, step count, last updated time, and actions such as Edit and Execute.
FR-03	Workflow Creation and Editing	The Create / Edit Workflow screen allows users to enter workflow name, description, workflow steps, step types, assigned agents, estimated time, and dependencies.

FR-04	Workflow Validation	The Workflow Editor screen provides a validation panel that shows overall validation status and validation checks.
FR-05	Execution Monitoring	The Executions screen displays execution statistics, workflow status, execution history, and execution status badges.
FR-06	Agent Management	The Agents screen displays available agents, agent types, current status, and active workflow count.
FR-07	Workflow Settings	The Settings screen allows users to configure notifications and auto-save behavior for workflow events.

3.2 Non-Functional Requirements

The prototype also reflects key non-functional requirements. The use of a consistent sidebar supports usability and navigation. Status badges improve visibility of workflow and execution states. Tables and cards make workflow data easier to scan. The validation panel supports reliability by showing whether a workflow is ready to save or publish. The agent and settings screens support extensibility because the system can later add more agent types or workflow configuration options.

3.3 Use Cases

The screen set covers the following use cases:

- **UC-01 View Workflow Dashboard:** user views workflow statistics and recent workflow activity.
- **UC-02 View Workflow List:** user views all available workflows.
- **UC-03 Create Workflow:** user creates a new workflow definition.
- **UC-04 Edit Workflow:** user modifies an existing workflow definition.
- **UC-05 Validate Workflow:** system checks workflow correctness before saving or publishing.
- **UC-06 Execute Workflow:** user starts workflow execution from the workflow list.
- **UC-07 Monitor Execution Status:** user monitors workflow execution progress and history.
- **UC-08 Manage Agents:** user views agents that can be assigned to workflow steps.
- **UC-09 Configure Workflow Settings:** user configures notification and auto-save preferences.

4 UI Design Objectives

The UI prototype follows these design objectives:

1. **Modern SaaS dashboard style.** The interface uses a clean sidebar, white cards, rounded corners, and simple tables to make the system look professional and easy to understand.
2. **Clear workflow status visibility.** Users should immediately understand whether a workflow is Running, Completed, Failed, or Draft.
3. **Efficient workflow management.** Users should be able to search workflows, create new workflows, edit existing workflows, and execute workflows from the main list.
4. **Structured workflow editing.** Workflow steps should be displayed in a clear table with order, step name, type, assigned agent, estimated time, and dependency.
5. **Visible validation feedback.** The validation panel should show whether the workflow is valid and which checks have passed.
6. **Execution monitoring support.** Users should be able to view execution statistics and execution history.
7. **Consistent navigation.** The sidebar provides access to Dashboard, Workflows, Workflow Editor, Executions, Agents, and Settings.

5 UI Flow Overview

The main UI flow starts when the user opens the Dashboard to view general workflow activity. The user can navigate to the Workflows screen to search and inspect workflow definitions. From the Workflows screen, the user can create a new workflow or edit an existing workflow. In the Workflow Editor, the user defines workflow information, configures steps, assigns agents, and validates the workflow. After the workflow is saved or published, the user can execute it and monitor execution through the Executions screen. The Agents screen supports agent selection, while the Settings screen supports workflow-related preferences.

Stage	User/System Action	Resulting UI State
Overview	User opens the module.	Dashboard screen with workflow statistics and recent activity.
Workflow Listing	User selects Workflows from the sidebar.	Workflow Management screen with searchable workflow table.
Creation / Editing	User clicks Create Workflow or Edit.	Create / Edit Workflow screen with workflow form and step table.
Validation	User clicks Validate Workflow.	Validation panel shows overall status and passed checks.
Execution	User clicks Execute from the workflow list.	Execution screen shows workflow execution history and status.
Agent Management	User opens Agents screen.	Agent list shows available agents and active workflow usage.
Configuration	User opens Settings screen.	Workflow-related settings and toggles are displayed.

6 UI Screen Descriptions

6.1 Dashboard Overview Screen

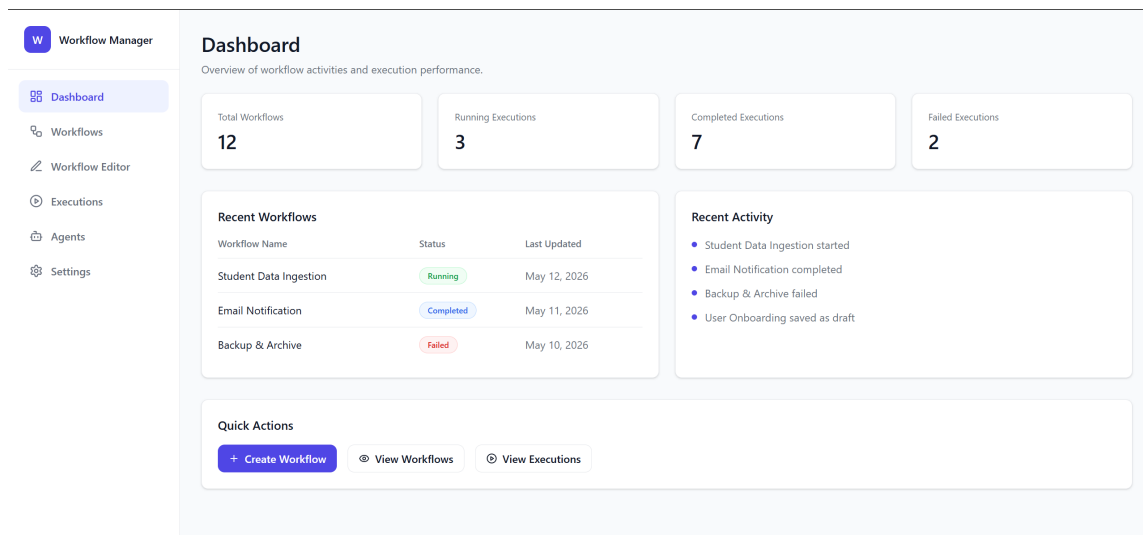


Figure 1: Dashboard Overview Screen

Purpose. This screen provides an overview of workflow activity and execution performance.

What it shows. The screen displays summary cards for total workflows, running executions, completed executions, and failed executions. It also shows a Recent Workflows table, a Recent Activity panel, and Quick Actions such as Create Workflow, View Workflows, and View Executions.

How users use it. The user uses this screen to quickly understand the current state of the workflow system. The user can inspect recent workflow activity and use the quick action buttons to move to workflow creation, workflow

list, or execution monitoring.

Related use cases. UC-01 View Workflow Dashboard; UC-02 View Workflow List; UC-03 Create Workflow; UC-07 Monitor Execution Status.

Design notes. The dashboard uses cards and a simple table layout to keep the overview easy to scan. It does not include complex charts because the main purpose is fast workflow status awareness.

6.2 Workflow Management Screen

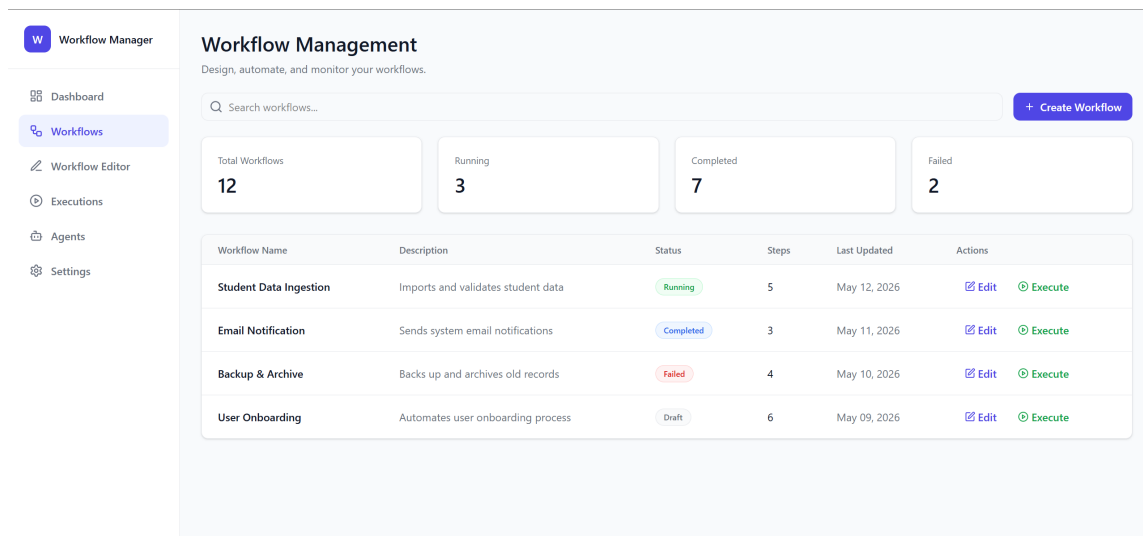


Figure 2: Workflow Management Screen

Purpose. This screen allows users to view, search, create, edit, and execute workflows.

What it shows. The screen displays a Workflow Management title, a search bar, a Create Workflow button, summary cards, and a workflow table. The table includes workflow name, description, status, number of steps, last updated date, and actions.

How users use it. The user searches for workflows, checks their status, and selects actions such as Edit or Execute. If the user wants to define a new process, they click Create Workflow. If the user wants to modify an existing workflow, they click Edit.

Related use cases. UC-02 View Workflow List; UC-03 Create Workflow; UC-04 Edit Workflow; UC-06 Execute Workflow.

Design notes. Status badges make workflow state visible without forcing the user to read long descriptions. The table design is suitable for workflow management because workflows are structured data items with repeated attributes.

6.3 Create / Edit Workflow Screen

Order	Step Name	Step Type	Assigned Agent	Estimated Time	Depends On
1	Import Student Data	Data Import	Data Import Agent	15 min	-
2	Validate Data	Validation	Validation Agent	20 min	1
3	Transform Data	Transformation	Transformation Agent	25 min	2

Figure 3: Create / Edit Workflow Screen

Purpose. This screen allows users to create or edit a workflow definition.

What it shows. The screen displays breadcrumb navigation, workflow name, description, workflow step table, Add Step button, and a validation panel. The step table includes order, step name, step type, assigned agent, estimated time, and dependency information.

How users use it. The user enters workflow information, adds or modifies workflow steps, assigns agents to each step, and checks dependencies. The user can validate the workflow, save it as a draft, or publish it when the structure is valid.

Related use cases. UC-03 Create Workflow; UC-04 Edit Workflow; UC-05 Validate Workflow; UC-08 Manage Agents.

Design notes. The two-column layout separates workflow editing from validation feedback. This allows the user to keep editing steps while still seeing whether the workflow is valid. The validation panel reduces errors before execution.

6.4 Executions Screen

Workflow	Status	Start Time	Duration
Student Data Ingestion	Running	May 12, 2026 09:30 AM	15 min
Email Notification	Completed	May 11, 2026 03:15 PM	5 min
Backup & Archive	Failed	May 10, 2026 11:45 AM	8 min

Figure 4: Executions Screen

Purpose. This screen allows users to monitor workflow execution history and execution status.

What it shows. The screen displays summary cards for total executions, running executions, completed executions, and failed executions. It also shows an execution table containing workflow name, execution status, start time, and duration.

How users use it. The user opens this screen to check which workflows are currently running, which ones have completed, and which ones have failed. The user can use this screen to track execution outcomes after starting a workflow.

Related use cases. UC-06 Execute Workflow; UC-07 Monitor Execution Status.

Design notes. The execution table focuses on operational monitoring rather than workflow design. It separates execution state from workflow definition, which keeps runtime information clearer.

6.5 Agent Management Screen

Agent Name	Type	Status	Active Workflows
Data Import Agent	Data Import	Active	5
Validation Agent	Validation	Active	3
Transformation Agent	Transformation	Active	4
Notification Agent	Notification	Inactive	2

Figure 5: Agent Management Screen

Purpose. This screen allows users to view and manage agents that can be assigned to workflow steps.

What it shows. The screen displays an agent table with agent name, type, status, and number of active workflows. Example agents include Data Import Agent, Validation Agent, Transformation Agent, and Notification Agent.

How users use it. The user reviews which agents are available and which agents are currently active or inactive. This helps the user understand which agents can be assigned when building workflow steps.

Related use cases. UC-08 Manage Agents; UC-03 Create Workflow; UC-04 Edit Workflow.

Design notes. The agent table supports the Workflow Editor screen because each workflow step requires an assigned agent. Keeping agent status visible helps prevent users from assigning inactive or unavailable agents.

6.6 Workflow Settings Screen

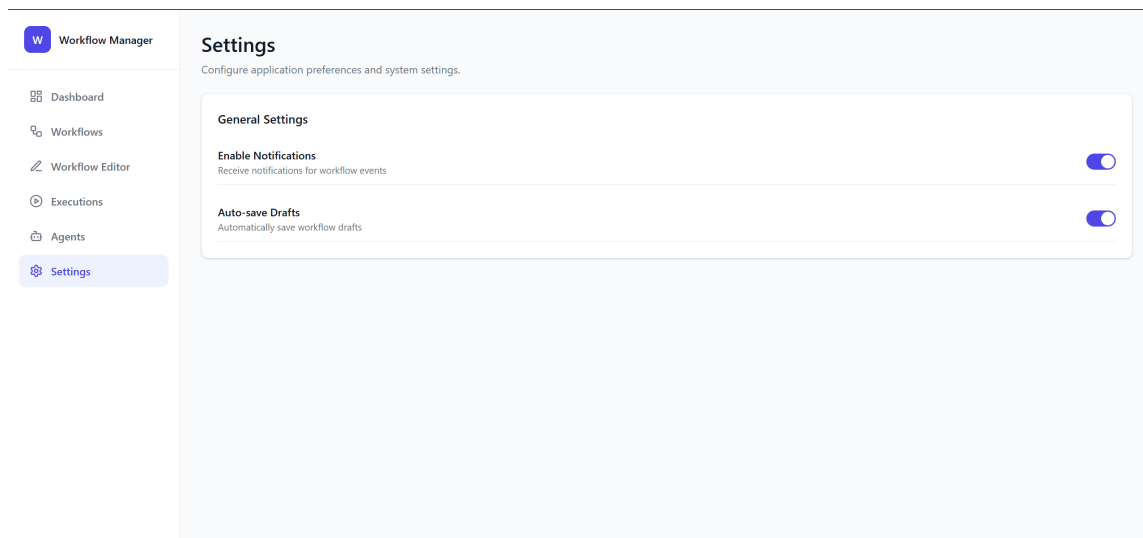


Figure 6: Workflow Settings Screen

Purpose. This screen allows users to configure workflow-related preferences.

What it shows. The screen displays general settings such as Enable Notifications and Auto-save Drafts. The settings are shown with clear labels, short descriptions, and toggle controls.

How users use it. The user turns workflow notifications on or off and enables or disables automatic draft saving. These options help users control how the workflow module behaves during editing and execution.

Related use cases. UC-09 Configure Workflow Settings.

Design notes. The settings screen uses a simple card layout because configuration options should be easy to understand and modify. Toggles are suitable because the current settings are binary preferences.

7 Traceability Matrix

UI Screen	Related Use Cases	Requirement Coverage
Dashboard Overview Screen	UC-01, UC-02, UC-03, UC-07	Shows workflow statistics, recent workflows, recent activity, and quick navigation actions.
Workflow Management Screen	UC-02, UC-03, UC-04, UC-06	Supports workflow listing, search, workflow creation, editing, and execution entry point.
Create / Edit Workflow Screen	UC-03, UC-04, UC-05, UC-08	Supports workflow definition, step configuration, agent assignment, dependency setup, and validation.
Executions Screen	UC-06, UC-07	Shows execution statistics, execution history, status badges, start time, and duration.
Agent Management Screen	UC-08, UC-03, UC-04	Shows available agents and supports agent awareness for workflow step assignment.
Workflow Settings Screen	UC-09	Supports workflow notification and auto-save configuration.

8 Evaluation Against UI Requirements

The UI prototype requirement asks for a main screen and at least two screens for key scenarios. This prototype satisfies the requirement by providing six screens: Dashboard, Workflow Management, Create / Edit Workflow, Executions, Agents, and Settings.

The main screen is the Workflow Management screen, where users can view workflows, search workflows, create new workflows, edit existing workflows, and execute workflows. The key scenario screens include Create / Edit Workflow and Executions. The Create / Edit Workflow screen supports workflow definition and validation, while the Executions screen supports execution monitoring.

The UI contains sufficient labels and controls for users to complete their tasks:

- A sidebar for navigating between Dashboard, Workflows, Workflow Editor, Executions, Agents, and Settings.
- A workflow list with status badges and action buttons.
- A workflow editor with workflow name, description, step table, assigned agents, dependencies, and validation panel.
- An executions screen with execution statistics and execution history.
- An agents screen showing agent availability and active workflow usage.
- A settings screen for notification and auto-save preferences.

The prototype also maintains consistency across screens through the same sidebar layout, color palette, card style, typography, and status badge design. This makes the workflow module easier to understand and keeps the interface aligned with a modern SaaS dashboard style.

9 Conclusion

The proposed UI prototype supports the main responsibilities of the Workflow Management module. It allows users to view workflow activity, manage workflow definitions, create and edit workflow steps, validate workflows, monitor executions, view agent availability, and configure workflow preferences.

The dashboard-style layout is suitable for a workflow management system because it presents structured operational information clearly. The prototype separates workflow definition from workflow execution, which improves readability and supports the main use cases of the module. Overall, the UI provides a clear and consistent foundation for implementing the Workflow Management feature in the virtual company workspace platform.